

FIN ST1902–ST1991

Automorphism-Natural Cell-Complex No-Go,
Refinement Conflict and the Hodge-Energy Frontier

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Six adaptive research rounds — 90 programmes

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This report tests the decisive target selected in Release 11.04: whether strict FIN weights and exact refinement uniquely generate an automorphism-natural oriented weighted 2-complex and gauge kinetic action. The result is a strong nonuniqueness theorem for support-only constructions and a conditional positive product-cell construction relative to the refinement map.

Abstract

ST1902–ST1991 analyse all automorphism-natural triangle face sets on the strict weighted support, classify natural face measures, test exact $12 \rightarrow 24$ refinement, and falsify the most canonical flag/Hodge repair.

The weighted strict graph has automorphism group exactly D_{12} : its unique maximum-weight distance-one edges form C_{12} , whose automorphisms are dihedral, and those preserve every cyclic distance class. The 220 triangles split into 12 D_{12} orbits of sizes

$$12, 24, 24, 24, 24, 12, 24, 24, 12, 12, 24, 4.$$

Therefore automorphism-natural face sets are arbitrary unions of these orbits, giving $2^{12} = 4096$ possibilities. Exact bit elimination proves that at least 2542 have full rank 55 over $\mathbb{F}_2$ and hence kill first homology over the rationals. At least 338 are inclusion-minimal, all using five orbits. Two different five-orbit complexes with 76 faces are independently verified by exact rational rank to have rank 55. Thus naturality, $H^1 = 0$, minimal orbit count and equal minimum face count do not select a complex.

Face measures remain nonunique after a face set is fixed. Positive D_{12} -natural weights form an $(\mathbb{R}_+)^{12}$ orbit cone, or $(\mathbb{R}_+)^5$ on a five-orbit minimal complex. Five explicit positive edge-derived rules—product, geometric mean, arithmetic mean, minimum and maximum—span rank five across the triangle orbits and are not proportional. Homogeneity does not remove the ambiguity: four of these rules have degree one. A discrete Hodge star and Wilson spectrum therefore depend on an unsourced face inner product and overall gauge coupling.

Refinement supplies a conditional cellular repair but sharpens the no-go. The support of $A_{24} = A_{12} \otimes I_2 + I_{12} \otimes B_q$ has 144 edges and cycle rank 121. Its flag complex contains 440 triangles with boundary rank 110, leaving

$$\dim H^1 = 11.$$

The typed product refinement supplies 66 edge-times-interval squares. Their addition raises the boundary rank to 121 and restores $H^1 = 0$. These squares are not cliques of the refined support. Hence the support-only flag functor is not product-refinement functorial; the product repair requires refinement history and retains six square-orbit weights plus all inherited base choices.

Positive Wilson actions exist for every positive face measure, but neither face areas nor action scale is sourced. A finite heat trace has spectral dimension zero at both asymptotic time extremes. Under iterated binary refinement,

$$Z_k(t) = Z_{12}(t) \prod_{\ell=1}^k (1 + e^{-2q_\ell t}),$$

so every intermediate spectral-dimension profile depends explicitly on the free sequence q_ℓ . A four-dimensional plateau can be fitted but is not derived. Complete-support nonlocal edges remain at every level, and the actions are Euclidean rather than Lorentzian.

The full clique complex is a canonical support-only object, but it is the 11-simplex with 4096 simplices. Its flag construction does not preserve the exact product refinement; the product Hodge repair is not support-only and still needs inner-product weights, cutoff scale and representations.

The decisive next theorem is therefore narrower and stronger than “find a 2-complex”: classify all positive face inner products for which gauge curvature energy commutes with FIN coarse/refinement maps over $12 \rightarrow 24 \rightarrow 48$, and determine whether they are unique up to one overall scale. Failure would be a direct no-go for strict gauge geometry without new assumptions.

Contents

1	Round I: ST1902–ST1916 — automorphisms and face sets	2
2	Round II: ST1917–ST1931 — face measures	2
3	Round III: ST1932–ST1946 — refinement conflict	3
4	Round IV: ST1947–ST1961 — Wilson action and continuum	3

5 Round V: ST1962–ST1976 — Hodge repair	4
6 Round VI: ST1977–ST1991 — decisive frontier	4
7 Most important result	4
8 Highest-value next theorem	5
9 Route ranking	5
10 Final interpretation	5
A Six-round ledger	6
B Reproducibility	6
C Nonclaims	6

1 Round I: ST1902–ST1916 — automorphisms and face sets

The six strict weights are distinct and the distance-one weight is uniquely maximal. Any weighted-graph automorphism must preserve the distance-one C_{12} subgraph, so it lies in D_{12} . Conversely every dihedral symmetry preserves cyclic distance and all weights. Therefore

$$\text{Aut}(A) = D_{12}, \quad |D_{12}| = 24.$$

The triangle orbits are characterized by cyclic side-distance triples. There are exactly twelve. A face set is automorphism-natural precisely when it is a union of complete orbits, yielding 4096 choices including the empty set and the full triangle set.

To test whether topology/minimality removes the ambiguity, every orbit subset was audited over $\mathbb{F}_2$ using exact bit Gaussian elimination. At least 2542 subsets have triangle-boundary rank 55. Because 55 is the maximum cycle rank, each also has full rational rank. At least 338 are inclusion-minimal, all using five triangle orbits and between 76 and 120 faces.

Two explicit examples use orbit sets

$$\{0, 1, 4, 5, 11\}, \text{quad}\{0, 2, 4, 5, 11\}.$$

Each has 76 faces and exact rational boundary rank 55, yet they are distinct.

Theorem 1 (automorphism/topology face-set no-go). *Weighted-graph naturality, vanishing H^1 , inclusion minimality, five-orbit support and minimum face count do not determine a unique strict 2-complex.*

The census count is a rigorous lower bound for rationally full-rank complexes: additional subsets could fail mod 2 while pass over the rationals. This only strengthens nonuniqueness.

2 Round II: ST1917–ST1931 — face measures

On all triangles, a positive D_{12} -natural measure is constant on each orbit:

$$\kappa \in (\mathbb{R}_+)^{12}.$$

A selected five-orbit minimal complex retains $(\mathbb{R}_+)^5$.

For a triangle with edge weights (w_1, w_2, w_3) , the following are all positive, symmetric and automorphism-natural:

$$w_1 w_2 w_3, \quad (w_1 w_2 w_3)^{1/3}, \quad \frac{w_1 + w_2 + w_3}{3}, \quad \min_i w_i, \quad \max_i w_i.$$

Their twelve orbit-value vectors have numerical rank five. The product and geometric-mean vectors are not proportional; their ratio varies by orbit. Geometric/arithmetic mean, minimum and maximum all scale linearly under $w \mapsto cw$, so degree-one homogeneity still leaves at least four candidates. More generally, arbitrary positive symmetric functions of two independent edge ratios preserve degree-one naturality.

The face measure defines the 2-form inner product/Hodge star and changes the gauge spectrum and EOM. Even after relative weights are fixed, an overall Wilson/gauge coupling remains free. Dimensionless edge weights do not provide face area, length squared or an action unit.

Maximum entropy requires a reference measure and constraints; a scalar Shannon or moment datum cannot select a vector in a 12-dimensional cone.

Theorem 2 (natural face-measure no-go). *Current strict edge weights plus positivity, symmetry and homogeneity do not determine a unique face measure or gauge kinetic normalization.*

3 Round III: ST1932–ST1946 — refinement conflict

For generic $q > 0$, the refined support is $K_{12} \square K_2$ with

$$V = 24, qquad E = 144, qquad b_1 = 121.$$

All 440 triangles lie entirely inside the two K_{12} layers. Exact mod-2 elimination gives triangle-boundary rank 110, so the flag complex has

$$\dim H^1 = 121 - 110 = 11.$$

The typed Cartesian product supplies one square for each of the 66 base edges:

$$(i, 0) \rightarrow (j, 0) \rightarrow (j, 1) \rightarrow (i, 1) \rightarrow (i, 0).$$

The square boundaries have rank 66, but add exactly 11 directions modulo the triangles. Together, triangles and squares have rank 121 and kill H^1 .

Theorem 3 (flag/refinement conflict). *The support-only flag complex does not preserve first-homology closure under the exact FIN product refinement. Restoring closure requires non-clique square cells obtained from the typed refinement map.*

Every natural base face set still lifts to both layers, so refinement does not choose among base candidates. The 66 squares split into six cyclic-distance orbits and retain a positive six-parameter weight cone. If q coincides with a base edge weight, fiber edges cannot even be identified by weight alone; refinement metadata is essential.

Thus a relative product-cell functor exists once the base complex and refinement morphism are provided. No absolute support-only functor is selected.

4 Round IV: ST1947–ST1961 — Wilson action and continuum

For positive face coefficients,

$$S_W = \sum_f \kappa_f (1 - \operatorname{Re} U_f) \geq 0.$$

For small $U_f = e^{iF_f}$,

$$1 - \cos F_f = \frac{1}{2} F_f^2 + O(F_f^4),$$

so every chosen positive measure supplies a consistent Euclidean quadratic curvature action. It does not fix face areas, lattice spacing, overall gauge coupling or action units.

For the finite base spectrum,

$$Z_{12}(t) = \sum_j e^{-t\lambda_j}$$

tends to 12 as $t \rightarrow 0$ and to 1 as $t \rightarrow \infty$. Its logarithmic spectral dimension tends to zero at both extremes; no finite graph has an asymptotic continuum power law.

Iterated binary refinements give the exact factorization

$$Z_k(t) = Z_{12}(t) \prod_{\ell=1}^k (1 + e^{-2q_\ell t}),$$

and

$$d_s(t) = 2t \left[\frac{\sum_j \lambda_j e^{-t\lambda_j}}{Z_{12}(t)} + \sum_\ell \frac{2q_\ell}{e^{2q_\ell t} + 1} \right].$$

The effective dimension depends on the entire free sequence q_ℓ . With constant q , it grows with layer count at fixed intermediate time. Geometric or other sequences can shape plateaux, including a fitted value near four, but that is a scaling assumption rather than a derivation.

Every refined vertex retains eleven complete-base neighbours. Product refinement adds fiber locality without removing base nonlocality. The action is Euclidean; Lorentzian signature and a causal cone remain absent.

5 Round V: ST1962–ST1976 — Hodge repair

The full flag complex of base K_{12} is canonical from support and equals the 11-simplex. It has $H^1 = 0$ and 4096 simplices including the empty simplex. This is mathematically canonical but not a 3+1-dimensional geometry.

At the refined level the flag complex contains triangles but not product squares, producing eleven one-form Hodge zero modes. The product-cell repair adds the squares and removes those modes. Therefore the two naturality principles conflict:

- *support naturality*: use the clique/flag complex of the current graph;
- *refinement naturality*: use the cellular product of the prior complex with the refinement interval.

The second uses morphism history, not only the current support.

Even after choosing product cells, a weighted Hodge Laplacian requires edge and face inner products and their relative scale. A spectral action further needs a cutoff function, dimensional cutoff and field representations.

Theorem 4 (support-only Hodge refinement no-go). *No support-only flag/Hodge construction in the declared class simultaneously reproduces base $H^1 = 0$ and exact product-refinement $H^1 = 0$ without adding non-clique cells from refinement metadata.*

6 Round VI: ST1977–ST1991 — decisive frontier

Object	Status	Boundary
weighted automorphism group D_{12}	[Proven]	strict graph
triangle orbit census	[Proven]	12 orbits
unique natural face set	[Refuted]	thousands of candidates
unique positive natural measure	[Refuted]	orbit cones/rule families
positive Wilson action	[Constructed conditional]	chosen cells/weights
flag complex	[Canonical support-only]	fails exact refinement closure
product cellular complex	[Conditional relative functor]	base/morphism supplied
spectral dimension four	[Not derived]	free q_ℓ /scale
continuum Maxwell physics	[Not established]	locality/units/signature absent

The updated cellular-refinement gate contains seven strict rows: base face selector, positive face measure, coherent refinement lift, energy-preserving Hodge compatibility, length/area/action scaling, local continuum/causal limit, and OA platform/record. The current strict repository passes zero complete rows. The relative product construction is partial conditional progress.

7 Most important result

The most important conclusion is:

no canonical gauge curvature geometry follows from strict support and basic naturality alone.

This remains true after imposing vanishing first homology, minimal orbit support, positive edge-derived measures, exact refinement and Hodge closure.

The constructive remnant is a *relative* cellular category: after choosing a base complex and retaining the explicit refinement morphism, lift base faces and add product squares/cubes. It is functorial relative to the history, but its initial complex and Hodge weights remain unsourced.

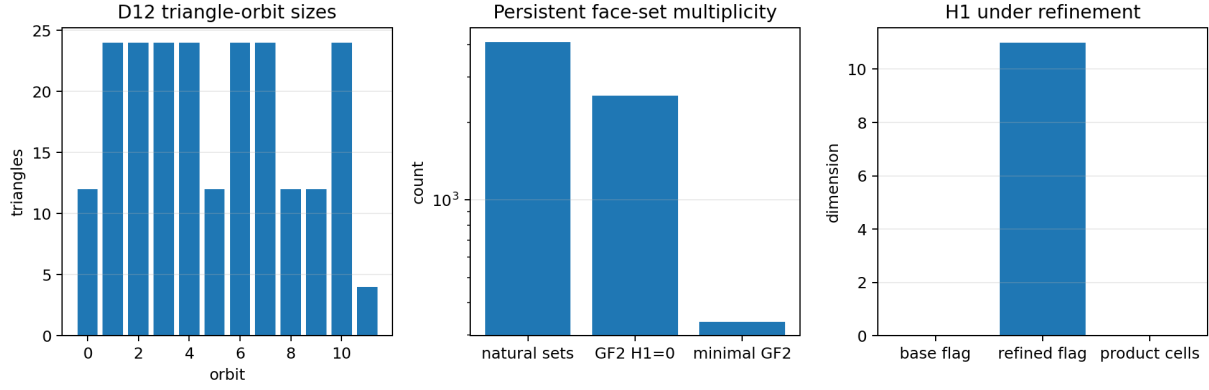


Figure 1: Left: the twelve triangle-orbit sizes. Centre: face-set multiplicity persists after topology/minimality gates. Right: the flag complex acquires eleven H^1 modes under refinement; product squares remove them.

8 Highest-value next theorem

The next problem with the greatest power to decide whether FIN can generate gauge physics is:

Classify all positive face inner products for which gauge-curvature energy commutes with FIN coarse/refinement maps over $12 \rightarrow 24 \rightarrow 48$, and prove either uniqueness up to one overall scale or a nonuniqueness theorem.

Required tests are existence, positivity, automorphism naturality, the $12 \rightarrow 24$ energy identity, $24 \rightarrow 48$ associativity, uniqueness modulo scale, and controlled continuum scaling. A failure would block strict gauge geometry without new assumptions. A unique solution would supply the first serious candidate for a strict Hodge/Wilson action.

9 Route ranking

Route	Rigorous-success estimate	Physics decision power
classify energy-preserving Hodge measures across 12–24–48	0.42	1.00
develop conditional product-cell $U(1)$ lattice model	0.82	0.50
support-only flag spectral action	0.15	0.55
continue APD/moment L_{total} repair	0.01	0.35
OA platform experiment	0.45	0.55

These are strategic judgments, not empirical probabilities.

10 Final interpretation

FIN currently supports a clean conditional gauge-matter action and a relative product-cell refinement, but not a unique strict gauge curvature geometry. Automorphism symmetry, topology, flag naturality and spectral dimension do not remove the choice of faces, Hodge weights or scaling. FIN therefore remains a finite Dirichlet–Markov–spectral information geometry with conditional gauge extensions, not a fundamental physical theory. The energy-preserving Hodge classification is now the sharpest mathematical decision point.

A Six-round ledger

Range	Target	Verdict
ST1902–ST1916	automorphism-natural faces	massive nonuniqueness
ST1917–ST1931	face measures	positive natural cone nonunique
ST1932–ST1946	refinement functor	flag/product conflict
ST1947–ST1961	Wilson/continuum	positive conditional action, dimension free
ST1962–ST1976	Hodge repair	support-only functor no-go
ST1977–ST1991	synthesis	energy-compatible Hodge frontier

B Reproducibility

The authoritative result sets run from `FIN_ST1902_ST1916_Results.json` through `FIN_ST1977_ST1991_Results.json`. Each contains fifteen programmes and packet SHA-256 references. The combined test `test_fin_st1902_st1991.py` verifies all ninety packets, orbit counts, exact ranks, measure rank, refinement homology, flag/Hodge obstruction, updated gate, figure and nonpromotion boundaries.

C Nonclaims

This report does not derive a base face set, Hodge measure, gauge coupling, lattice spacing, continuum dimension, Lorentz signature, Maxwell normalization, state ontology, apparatus or evidence. It does not complete legacy role transfer, QW-2191, Standard Model, gravity, L_{total} or Theory of Everything. All future FIN PDFs remain English.

References

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